Database system structure

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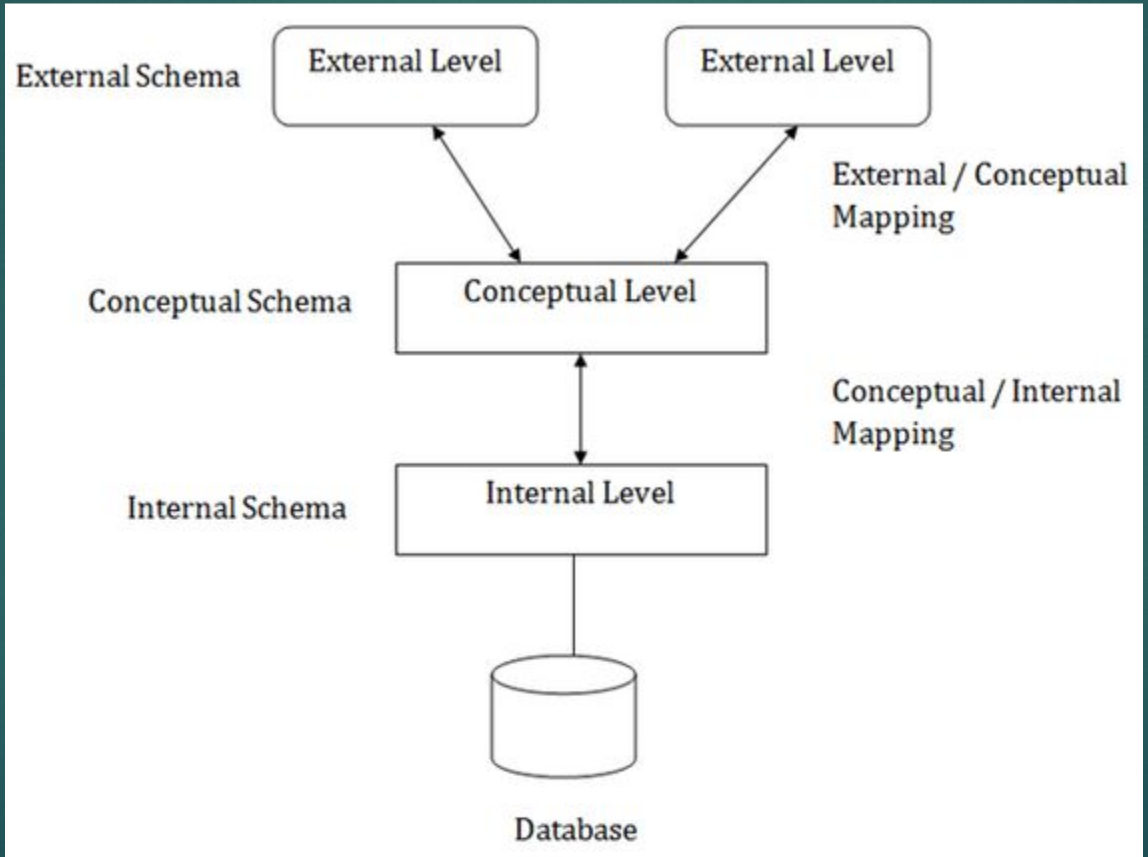
Three schema Architecture

- The three schema architecture is also called ANSI/SPARC architecture or three-level architecture.
- This framework is used to describe the structure of a specific database system.
- The three schema architecture is also used to separate the user applications and physical database.
- The three schema architecture contains three-levels. It breaks the database down into three different categories.

The three-schema architecture



The 3-tier architecture

- The 3-tier architecture is a commonly used architectural approach in Database Management Systems (DBMSs) for the design and development of applications that work with databases.
- The 3-tier architecture divides an application's components into three tiers or layers. Each layer has its own set of responsibilities.
- DBMS 3-Tier architecture divides the complete system into three inter-related but independent modules as shown below.
- **Types of Schema:** The database has several schema based on the levels of abstraction.
- (1) Physical Schema: The physical schema is a database design described at the physical level of abstraction.
- (2) Logical Schema: The logical schema is a database design at the logical level of abstraction.
- (3) Subschema:A database may have several views at the view level which are called subschemas.

Physical Level

- At the physical level, the information about the location of database objects in the data store is kept.
- Various users of DBMS are unaware of the locations of these objects.
- In simple terms, physical level of a database describes how the data is being stored in secondary storage devices like disks and tapes and also gives insights on additional storage details.
- The physical or internal layer is the lowest level of data abstraction in the database management system.

Conceptual Level

- At conceptual level, data is represented in the form of various database tables.
- For Example, STUDENT database may contain STUDENT and COURSE tables which will be visible to users but users are unaware of their storage.
- Also referred as logical schema,it describes what kind of data is to be stored in the database.
- The logical or conceptual level is the intermediate or next level of data abstraction. It explains what data is going to be stored in the database and what the relationship is between them.

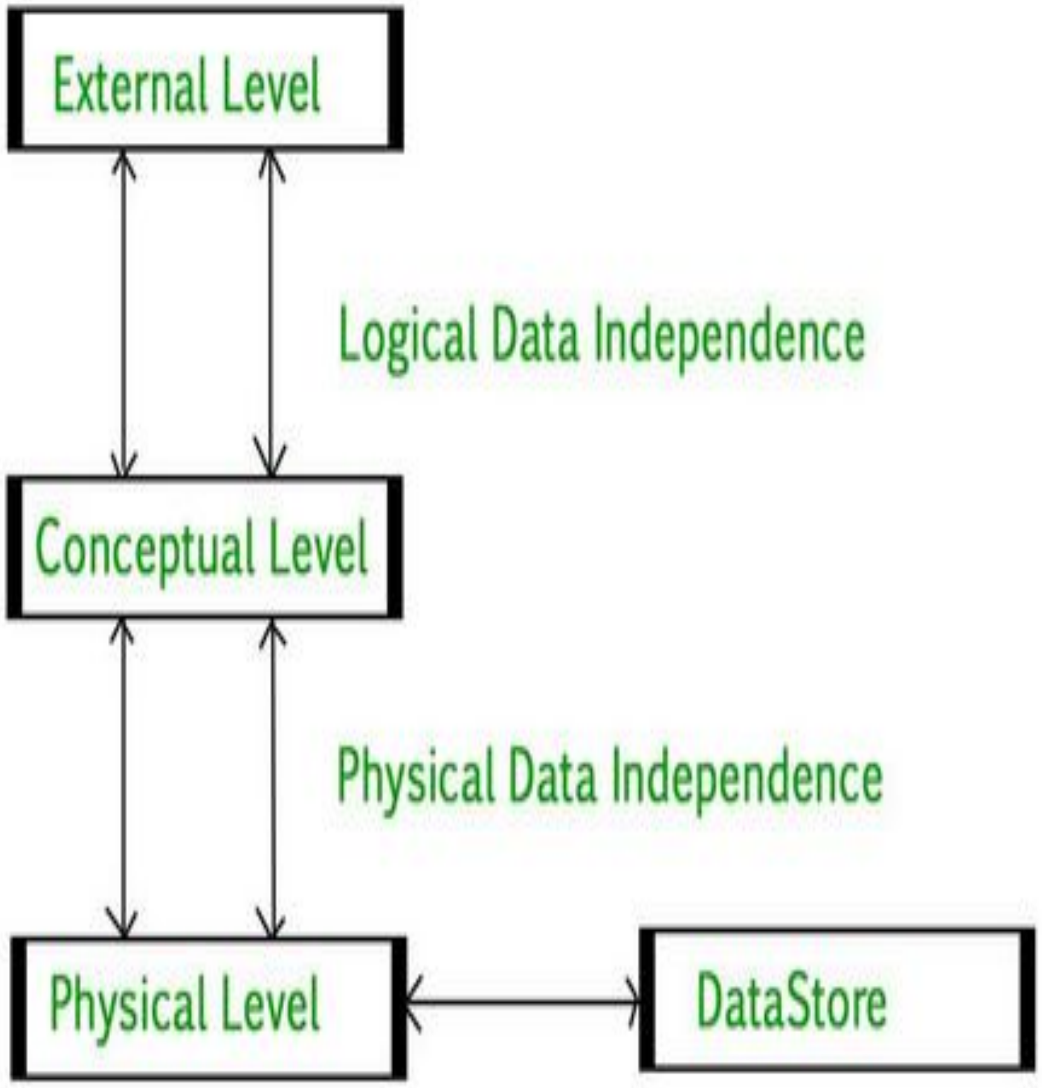
External Level

- An external level specifies a view of the data in terms of conceptual level tables. Each external level view is used to cater to the needs of a particular category of users.
- For Example, FACULTY of a university is interested in looking course details of students, STUDENTS are interested in looking at all details related to academics, accounts, courses and hostel details as well.
- So, different views can be generated for different users. The main focus of external level is data abstraction.

Data Independence

Data independence means a chang of data at one level should not affect another level. Two types of data independence are present in this architecture:

Physical Data Independence: Any change in the physical location of tables and indexes should not affect the conceptual level or external view data. This data independence is easy to achieve and implemented by most of the DBMS.



DBMS 3-tier architecture

Conceptual Data Independence

- The data at conceptual level schema and external level schema must be independent.
- This means a change in conceptual schema should not affect external schema. e.g.; Adding or deleting attributes of a table should not affect the user's view of the table.
- But this type of independence is difficult to achieve as compared to physical data independence because the changes in conceptual schema are reflected in the user's view.

Phases of Database Design

- Database designing for a real-world application starts from capturing the requirements to physical implementation using DBMS software which consists of following steps shown below.
- **Conceptual Design:** The requirements of database are captured using high level conceptual data model. For Example, the ER model is used for the conceptual design of the database.
- **Logical Design:** Logical Design represents data in the form of relational model. ER diagram produced in the conceptual design phase is used to convert the data into the Relational Model.
- **Physical Design:** In physical design, data in relational model is implemented using commercial DBMS like Oracle, DB2.

Instances and Schemas

- Schema: The overall design of the database is called schema

- For example - In a program we do variable declaration and assignment of values to the variable. The variable declaration is called schema and the value assigned to the variable is called instance. The schema for the student record can be

RollNo	Name	Marks

Instances:

- **Instances:** When information is inserted or deleted from the database then the database gets changed. The collection of information at particular moment is called instances. For example - following is an instance of student database

RollNo	Name	Marks
10	AAA	43
20	BBB	67

What are the Data Models in DBMS?

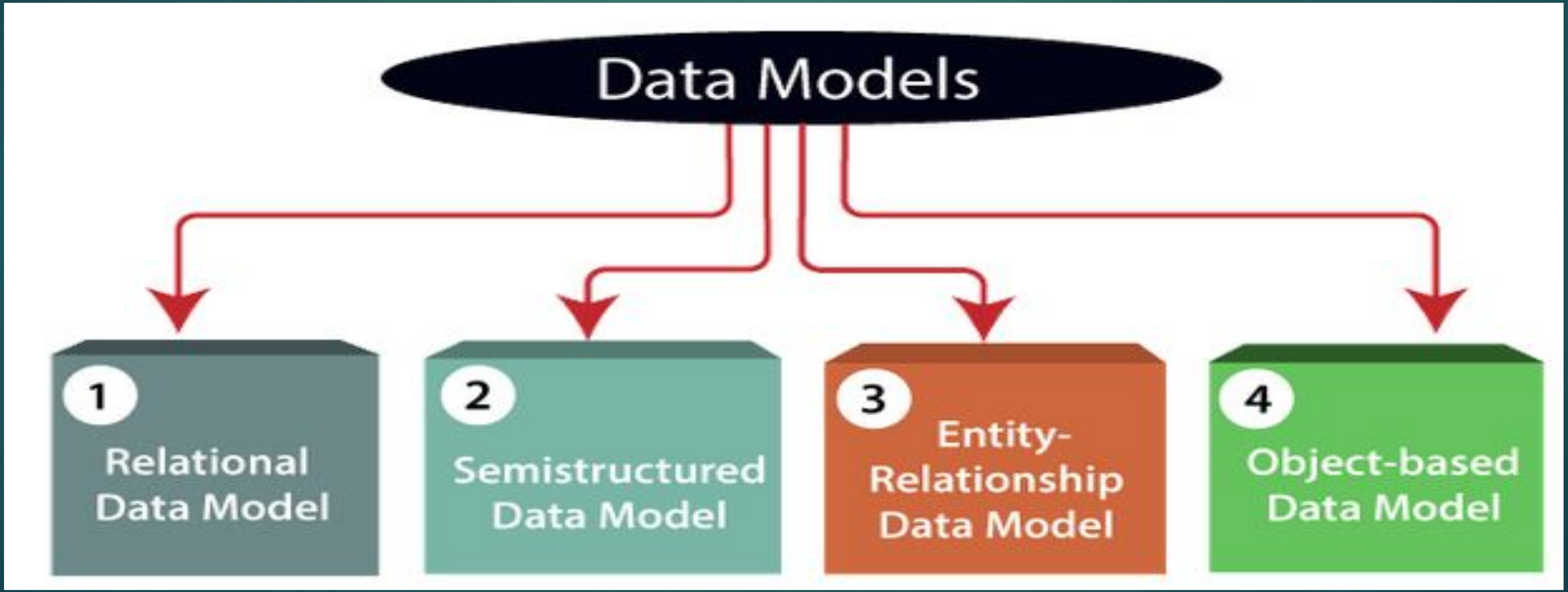
- The Data Model gives us an idea of how the final system would look after it has been fully implemented. It specifies the data items as well as the relationships between them.
- In a database management system, data models are often used to show how data is connected, stored, accessed, and changed.
- We portray the information using a set of symbols and language so that members of an organisation may understand and comprehend it and then communicate.

Types of Data Models in DBMS

- Though there are other data models in use today, the Relational model is the most used. Aside from the relational model, there are a variety of different data models
- Hierarchical Model
- Network Model
- Entity-Relationship Model
- Relational Model
- Object-Oriented Data Model
- Object-Relational Data Model
- Flat Data Model
- Semi-Structured Data Model
- Associative Data Model
- Context Data Model

Data Models

- Data Model is the modeling of the data description, data semantics, and consistency constraints of the data.



Hierarchical Model

- The hierarchical Model is one of the oldest models in the data model which was developed by IBM, in the 1950s.
- In a hierarchical model, data are viewed as a collection of tables, or we can say segments that form a hierarchical relation.
- In this, the data is organized into a tree-like structure where each record consists of one parent record and many children.
- Even if the segments are connected as a chain-like structure by logical associations, then the instant structure can be a fan structure with multiple branches. We call the illogical associations as directional associations.

Network Model

- The Network Model was formalized by the Database Task group in the 1960s.
- This model is the generalization of the hierarchical model.
- This model can consist of multiple parent segments and these segments are grouped as levels but there exists a logical association between the segments belonging to any level.
- Mostly, there exists a many-to-many logical association between any of the two segments.

Relational Data Model:

- This type of model designs the data in the form of rows and columns within a table. Thus, a relational model uses tables for representing data and in-between relationships.
- Tables are also called relations. This model was initially described by Edgar F. Codd, in 1969. The relational data model is the widely used model which is primarily used by commercial data processing applications.

Entity-Relationship Data Model:

- An ER model is the logical representation of data as objects and relationships among them.
- These objects are known as entities, and relationship is an association among these entities.
- This model was designed by Peter Chen and published in 1976 papers. It was widely used in database designing. A set of attributes describe the entities.
- For example, student_name, student_id describes the 'student' entity. A set of the same type of entities is known as an 'Entity set', and the set of the same type of relationships is known as 'relationship set'.

Object-based Data Model:

- An extension of the ER model with notions of functions, encapsulation, and object identity, as well.
- This model supports a rich type system that includes structured and collection types.
- Thus, in 1980s, various database systems following the object-oriented approach were developed. Here, the objects are nothing but the data carrying its properties.

Semistructured Data Model:

- This type of data model is different from the other three data models (explained above).
- The semi structured data model allows the data specifications at places where the individual data items of the same type may have different attributes sets.
- Semi-Structured data models deal with the data in a flexible way. Some entities may have extra attributes and some entities may have some missing attributes. Basically, you can represent data here in a flexible way.
- The Extensible Markup Language, also known as XML, is widely used for representing the semi structured data. it gains importance because of its application in the exchange of data.

Associative Data Model and Context Data Model

- It is a model in which the data is separated into two sections. Everything that has its own existence is referred to as an entity, and the relationships between these entities are referred to as associations.
- Items and links are two types of data that are separated into two components.
- **Context Data Model**
- The Context Data Model is made up of various models. This includes models such as network models and relational models, among others.

Flat Database model

- database has to be represented as a single table, with all records saved as single rows of data separated by tabs or commas.
- A flat database refers to a simple DB system in which each database has to be represented as a single table, with all records saved as single rows of data separated by tabs or commas. A simple text file is typically used to store and physically display the table.
- Flat databases are not appropriate for most software applications that need the representation and storage of complex business interactions due to their constraints. However, some application developers continue to use flat files to save money and time when connecting a relational database.
- Flat databases, or flat-file databases, are another name for them.

Advantages of Data Models

- Data Models help us in representing data accurately.
- It helps us in finding the missing data and also in minimizing Data Redundancy.
- Data Model provides data security in a better way.
- The data model should be detailed enough to be used for building the physical database.
- The information in the data model can be used for defining the relationship between tables, primary and foreign keys, and stored procedures.